

CV funk

CV funk Sands Collection for VCV Rack



User Manual

Version 2.3.8

CV funk

LICENSES

End User License Agreement for Software Sold on VCV Rack.com

—

1. USE

This software is licensed, not sold.

The purchase of this software by a user grants the user one license for the software. A license grants use on all operating systems on all computers owned by the user, provided the primary user is the license holder.

Businesses and organizations must purchase a license for each user of the software.

You must not remove or alter any copyright notices on any copy of this software.

You must not disassemble, decompile, or reverse engineer the software, in whole or in part.

You must not distribute graphics and resources from this software package.

2. DISTRIBUTION

You must not distribute or sell licenses of this software to third parties. Licenses are revoked if distributed or sold to third parties.

3. WARRANTIES

THIS SOFTWARE IS PROVIDED BY THE COPYRIGHT HOLDERS AND CONTRIBUTORS "AS IS" AND ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE DISCLAIMED. IN NO EVENT SHALL THE COPYRIGHT HOLDER BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

CV funk Sands

The **CVfunk Sands Collection** is composed of nine original modules that work as a system. The core workflow connects *Drifts* as a CV generator, *Mirage* as a chord-aware quantizer with unique oscillator voices, *Zephyr* as a morphable oscillator, *Strands* a complex physical modeling voice, and *Droplet* providing physical modeling drums and reverb. *Dunes*, *Crosswinds*, *Arcs*, and *Sirocco* extend the system with function generation, rhythmic displacement, probabilistic routing, and a generative 'ecology' for CV.

Mirage 5

A lead quantizer that expands a chord into a full scale by scoring passing notes based on the chord's harmonic character.

Crosswinds 7

16-step rhythmic gate displacement sequencer with per-step offset, ratcheting, and an animated waveform display.

Drifts 9

Four-layer clock-synced oscillator with morphable waveshapes, harmonic ratios, and a summed CV output. Also capable of audio-rate operation.

Arcs 11

Probabilistic 8x8 signal router with crossfade, crosstalk control, and an envelope output.

Dunes 13

Dual function generator with GROW/DECAY/WAIT phases, analog logic outputs, and a central VU/mixer section.

Sirocco 15

Desert ecosystem simulator: evolving grass and locust populations drive CV outputs for wind, rain, and spatial movement.

- ### Zephyr** 17
Polyphonic V/Oct oscillator and LFO with three
morphable wave families, continuous 2D waveshape
morphing, FM inputs, and hard/soft sync.
- ### Droplet** 19
Polyphonic Karplus-Strong 8-node resonator ring
feeding an 8-node plate reverb, with a morphable
strike excitation and dynamics-driven modulation
bus.
- ### Strands**
Hurdy-gurdy physical model with six bowed strings,
twelve sympathetic resonators, trompette chien, and
true stereo dual-mono signal path.

MIRAGE

ROOT

Root selector, knob or CV. Sets the tonal center for passing note calculation. The module ranks suggested roots by how harmonious they are with the set notes.

CHORD IN

A poly V/oct cable defines the active chord tones. If nothing is connected, use the on-panel keyboard widget to set the chord manually.

CV IN

Lead melody input, V/oct. The signal to quantize

TENSION / LEAD

TENSION always outputs the full extended scale regardless of gate state. Useful for a second voice that always moves through passing notes even when the LEAD output is constrained.



DISPLAY

Bright gold dots mark chord tones. Dimmer dots mark active passing notes. When the gate is closed, passing note dots go grey. Interval name labels appear below the top-priority passing notes.

MODE

Select between 12Tet, Just Intervals, Pythagorean Intervals, or 14Tet.

EXPRESSION

Increase the number of passing notes in-between the defined notes, based on the quantization MODE. These notes are added in order of harmonicity with the selected ROOT.

DELTA

Difference between TENSION and LEAD.

BEND

Microtonal offset of TENSION OUT from the nearest 12TET semitone.

VELOCITY

Velocity output, 0–10V. Chord tones output 10V. Passing notes output a graded voltage from 1V to 9V depending on harmonic priority.

MIRAGE is a chord-aware quantizer with expression-controlled passing notes, microtonal modes, and velocity output. The module analyzes the set notes and ranks potential root notes to choose which passing notes to prefer.

MIRAGE

Tuning Modes

- **12TET:** Equal temperament, chromatic semitones.
- **Just:** Harmonic series ratios from the root, passing notes from just intervals (7/4, 5/4, 11/8, etc.). DELTA carries deviation from 12TET – patch to timbre of oscillators.
- **Pythagorean:** Built from stacked pure fifths (3/2); passing notes drawn from 3-limit ratios (9/8, 81/64, 32/27, etc.) for bright thirds and Pythagorean-comma color.
- **24TET:** Quarter-tone equal temperament, 12 extra positions between semitones. DELTA outputs $\pm 1/24$ V on quarter-tones.

Usage

1. **Chord:** Set the chord on the panel keyboard, or patch a poly V/oct to POLY IN.
2. **Root:** Set ROOT to match your tonal center.
3. **Input:** Patch a melodic CV source to CV IN. A slow LFO, sequencer, or *Drifts* MIX OUT all work well.
4. **Output:** Take CV OUT to an oscillator or other pitch destination.
5. **Expression:** Adjust EXPRESSION to reveal passing notes. Start at zero for chord-tone-only quantization, then increase to add passing color.
6. **Gate:** Optionally patch a gate to GATE IN to control when passing notes are accessible. Useful for constraining fills to chord tones during cadences.
7. **Sample-hold:** Patch a trigger to TRIG IN if you want all outputs to hold between triggers.

Patching Ideas

1. **Melodic lead over chord changes:** Patch *Drifts* MIX OUT to CV IN. Connect a poly chord sequencer to POLY IN. As the chord changes, the scale updates and the lead stays harmonically correct throughout.
2. **Microtonal inflection:** Patch TENSION OUT to a second oscillator and DELTA OUT to that oscillator's FM input. The second voice follows the just scale with natural intonation for a subtly choir-like effect.
3. **Velocity-aware dynamics:** Patch VEL OUT to an envelope amount or VCA level. Chord tones hit at full strength and passing notes are softer, giving melodic lines natural phrasing without extra envelopes.
4. **Expression sweep:** Patch an envelope or LFO into the EXPRESSION CV input. The scale opens up at phrase peaks and closes back to chord tones in between.

CROSSWINDS

DISPLAY

Live updating display of the entire displaced gate pattern across all active steps. Each gate event appears as a pulse at its displaced position within the beat. A moving cursor shows real-time playback position.

RHYTHMIC DISPLACEMENT

Per-step displacement. Clockwise shifts the gate later in the beat. Counter-clockwise shifts it earlier. Range is -1 to +1 of one step width, scaled by DEPTH.

Enable or disable each step. Lit means active. Can be toggled while running.

RATCHETING

A ratchet count of N fires N evenly-spaced gates within that step's beat.

KNOB MODE BUTTON

In RATCHET mode (KNOB MODE button lit), trimpots set the ratchet count for each step instead of displacement offset.

ON / OFF

Start/stop the sequencer.



LENGTH

Set the length of the sequence.

DEPTH

Set the depth of the rhythmic displacement.

CV MAP

The MAP CV input accepts a 16-channel poly cable. Each channel corresponds to a step. The context menu sets what the CV drives: Ratchets (count 1-8 per step), Offsets (replaces knob values per step), or Step Gates (high means step active, low means step inactive).

BEAT

Clock input

RESET

Resets the sequencer to the start.

END / OUT

END Fires a gate on each sequence loop, or at the end of a one-shot pass.

OUT is the displaced gate output.

CROSSWINDS is 16-step rhythmic gate displacement sequencer with per-step offset, ratcheting, and an animated waveform display.

CROSSWINDS

Usage

1. **Clock:** Patch a clock to the BEAT jack.
2. **Steps:** Enable the steps you want using the step buttons.
3. **Offsets:** Dial the OFFSET knobs. Positive values push the gate later. Negative values pull it earlier.
4. **Depth:** Use DEPTH to scale the total displacement range in or out.
5. **Length:** Set LEN to control how many steps loop.
6. **Display:** Watch the waveform display to understand the resulting gate pattern.
7. **Transport:** Use ON/OFF to start and stop. Use RESET to re-align to beat zero.

Patching Ideas

1. **Swing:** Set most offsets to zero. Nudge beats 2 and 4 slightly positive for classic swing.
2. **Polyrhythmic displacement:** Set strongly different offsets on alternating steps for complex rhythmic displacement at a consistent tempo.
3. **Ratchet accents:** Set MAP CV Destination to Ratchets. Send an envelope or LFO on selected poly channels to add fills on specific steps that respond to patch energy.
4. **One-shot fills:** Enable one-shot mode. Patch END OUT back to ON/OFF to retrigger a fill pattern on cue from another source.
5. **Humanization:** Patch a slow LFO into the DEPTH CV input to make the overall displacement drift in and out over time.

Context Menu

1. **One-shot mode:** The sequence plays once then stops. A pulse to ON/OFF restarts it.
2. **Sync to clock on start/reset:** When enabled, the module waits for the next incoming clock pulse before advancing after a start or reset instead of free-running immediately.
3. **Gate Width:** Sets gate output length from 1 ms (trigger) up to 99% of one beat (sustained gate).
4. **MAP CV Destination:** Selects what the 16-channel poly CV controls. Options are Ratchets, Offsets, or Step Gates.
5. **Step Gate CV Mode:** Toggle (rising edge flips step state) or Direct Set (high means on, low means off).

DRIFTS

DISPLAY

Live updating display of the entire displaced gate pattern across all active steps. Each gate event appears as a pulse at its displaced position within the beat. A moving cursor shows real-time playback position.

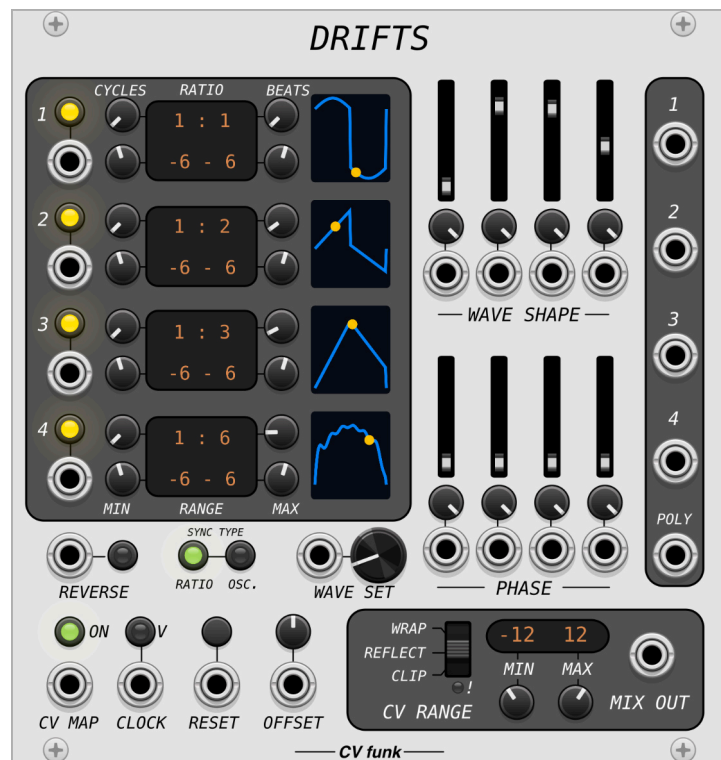
RHYTHMIC DISPLACEMENT

Per-step displacement. Clockwise shifts the gate later in the beat. Counter-clockwise shifts it earlier. Range is -1 to +1 of one step width, scaled by DEPTH.

Enable or disable each step. Lit means active. Can be toggled while running.

RATCHETING

A ratchet count of N fires N evenly-spaced gates within that step's beat.



KNOB MODE BUTTON

In RATCHET mode (KNOB MODE button lit), trimpots set the ratchet count for each step instead of displacement offset.

ON / OFF

Start/stop the sequencer.

BEAT

Clock input

RESET

Resets the sequencer to the start.

BEAT

Clock input

RESET

Resets the sequencer to the start.

DRIFTS is a four-layer clock-synced oscillator with morphable waveshapes, harmonic ratios, and a summed CV output. Also capable of audio-rate operation.

DRIFTS

Usage